

Study Planner - UI/UX Audit & Redesign Proposals

Evidence-based review for university students struggling with academics

September 2026 - Prepared from a full review of templates, CSS, and JavaScript

1. Executive Summary

Study Planner has an unusually strong design foundation: a disciplined token system, semantic status colors that never borrow the brand accent, global focus rings, a real motion scale, reduced-motion support, and an explicit stacking hierarchy. The dashboard's stated mission - 'a clear next step makes a focused study day' - is exactly right for the audience.

The problem is that most of the app contradicts that mission. A sturdily built system is being asked to serve the most fragile user in software: a university student who is anxious, behind, and low on self-efficacy. That user needs calm, momentum, and never to be punished for a mistake. The app currently delivers overload, dead-ends, and phantom data at exactly the moments of maximum stress. Most of the fixes proposed here are trust repairs, not visual polish.

2. What Is Already Strong

- **Design tokens and semantics:** one brand color, a small fixed semantic set (status/priority never borrow the accent), a 3-step elevation hierarchy, and a shared motion scale (variables.css).
- **Accessibility foundations:** focus-visible rings are set globally and never stripped (base.css), reduced-motion is honored via a settings toggle and prefers-reduced-motion, the course sidebar is keyboard-operable with roving arrows.
- **Feedback in the right places:** the AI chat has a typing indicator, disables Send while streaming, and the collab task panel has skeleton loading states and optimistic updates with rollback.
- **Empty states are mostly helpful and well-written** (courses, workspaces, AI landing, subject materials).
- **The dashboard hero states the right mental model for the audience.**

3. Core Diagnosis

The app is build-complete but not fully 'designed' in the emotional sense. It treats the student like a power user who enjoys administration, when the target user is drowning. Four systemic problems follow from this and shape everything below:

- **Phantom features are presented as real.**

Skeleton UIs (Grades, per-course Calendar, weekly chart) imply capability that does not exist, so students plan around data that will never arrive.

- **The user is punished for mistakes.**

No edit/delete for core objects that were typed by hand, instant destructive actions with no undo, and a timed quiz that silently submits.

- **Dead ends at maximum anxiety.**

The practice-quiz result page has no forward links; the low-confidence review loop routes to an invite-code gate; the Report and the full AI Hub are orphaned.

- **Cognitive load is pushed onto the student.**

One screen hosts three add-forms and four competing CTAs; the same object has four names; quiz entry uses two nearly identical pages.

4. Findings - Tier 1: Trust-Breakers

1. **Phantom features on the pages students plan around.**

The Class Calendar tab is a hardcoded 'Mon - Lecture / Wed - Tutorial / Fri - Quiz' (subject.html). The Grades tab renders 'Midterm - / Assignments - / Final exam -' with weights but no input path. 'Weekly study minutes' on Progress is a heading and a platitude with no chart (progress.html). Build these, badge them 'coming soon', or remove the skeleton.

2. **The recovery loop is a trap.**

Progress 'Needs review' routes to quiz_take_page - 'Enter a quiz invite code shared by a classmate' (progress.html -> quiz_take.html). The student flagged for low confidence lands on an unrelated entry screen with nothing tied to their flagged topics.

3. The Report and the full AI Hub are invisible.

report.html links only to itself; /ai-hub appears in no navigation, command palette, or profile menu. Two flagship learning features are unreachable without typing the URL.

4. The AI quick action sends a broken prompt.

The empty-state card 'Explain a topic' substitutes {subject} with an empty string (ai_hub.js vs app.py), so it sends 'Explain to me like I'm a beginner' - the hero affordance of the hero feature is broken.

5. Findings - Tier 2: Punishment and No-Undo

5. Course mistakes are permanent.

No edit or delete exists for courses anywhere (no routes). A mistyped course code, title, or lecturer can never be corrected.

6. Deadlines cannot be deleted. Notes cannot be edited.

Deadlines only support mark-done (no reopen, no delete). A backend route notes_update exists but no UI calls it - a typo means delete and recreate.

7. Confirmation policy is inconsistent.

Calendar, notes, and session deletes confirm; milestone delete, member remove, dependency remove, AI conversation delete, and 'Clear conversation' fire instantly (ai_hub.js, _panel.html).

8. Timed quiz auto-submits with no warning, then dead-ends.

At 0:00 the form submits unanswered questions silently (personal-quiz.js), and the result page has no links at all - no retake, no back to subject, no next step.

6. Findings - Tier 3: Cognitive Load and Navigation

9. Schedule is the heaviest screen in the app.

Three full add-forms and three tables on one page, with four competing primary CTAs: Start a focus session, Add Event, Add Task, Add Deadline. A student arriving to check 'what is due' must navigate a form farm.

10. Terminology drift across every surface.

'Courses', 'Subjects', and 'Semester' are used interchangeably across home, courses, the per-course sidebar, onboarding, and About. The same object has four names; users cannot tell whether they are different things.

11. Two near-identical 'Take a Quiz' screens do different jobs.

An invite-code gate and the actual attempt page are confusingly similar with no distinguishing affordance; a first attempt is a wrong door.

12. Settings requires five separate save submissions.

Five save buttons for profile, notifications, study, appearance, and privacy; every tiny change is a form submission with form-level confirmation. Confirmation fatigue is real for people already avoiding work.

7. Findings - Tier 4: Feedback, Brokenness, Accessibility

13. Group chat/comments break after one message.

hci.js disables the submit button on every form and never re-enables it; the AJAX handlers only call form.reset(). The first comment permanently kills the Send button until reload.

14. Skip link is a no-op on core pages.

subject.html, quiz.html, and quiz_results.html override the content block, dropping the id=main-content target from base.html. Screen-reader users get a skip link that focuses nothing.

15. Accessibility gaps.

Welcome-slide dots are focusable buttons inside an aria-hidden subtree with a color-only active state; the AI drawer and command palette are modals with no focus trap; AI modes use role=tab without a tablist; AI status errors sit outside a live region; the file-upload input has no label; the session page asks for a raw numeric 'Task id' and its confidence radios have no group semantics.

16. Mobile is an afterthought.

The 8-column deadlines table sits in a bare card with no overflow wrapper - the primary phone view squeezes or breaks the layout.

8. Proposals - Immediate Fixes (Week 1)

- Fix hci.js so form-disabling is opt-out for AJAX forms, or re-enable the button when the fetch completes. Group study is currently broken at message number two.

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Fix the {subject} placeholder to send the real active course subject, or ask the student once.

- Re-add id=main-content to the main block so the skip link works on every page.
- Add course edit/delete and deadline delete routes plus UI - small change, high relief.
- Wire the Report and the AI Hub into the navigation and command palette, or remove the Report.
- Give the timed quiz a 30-second warning and a 5-second 'submit anyway' gateway, and add forward links to the result page (retake, back to subject, AI review).
- Wrap wide tables in an overflow-x container and set min-widths so mobile never breaks.

9. Proposals - The 'No Phantom Features' Pass (Weeks 2-3)

- Implement Grades and the per-course calendar from real data, or label them 'Coming soon' and remove the skeletons. Fake features are worse than no features: students anchor on them.
- Term kill: pick one word - 'Courses' - and apply it everywhere, including onboarding and About copy.
- Reduce Schedule to one primary action; move the three add-forms into a modal or a single '+ Add' menu; tab the deadlines list (To do / This week / Later).
- Collapse Settings to one or two save forms: autosave changes on edit (debounced) with a single 'Save all' fallback.

10. Proposals - 'Protect the Fragile Student' Pass (Weeks 3-4)

- One clear next step per screen, starting with the dashboard: keep the hero, move everything else into tabs (Today / This Week / Courses). If it is not a decision for right now, it does not belong on the landing screen.
- Repair the review loop: Progress 'Needs review' deep-links to a specific course and its own materials, not the invite gate, and remembers what the student got wrong.
- Add a '20-minute day' mode: when only 20 minutes are available, propose the single highest-leverage 20-minute task. Struggling students think all-or-nothing; give them the escape hatch.
- Soft language pass: 'You are behind on...' becomes 'We will rebuild this into a smaller, doable plan.' The job of this app is momentum, not judgment.

11. Proposals - Differentiators (Next Quarter)

- Get-Unstuck AI grounded in the student's own materials - a lightweight RAG layer over notes and uploads, with answers that cite their own notes instead of generic chatter (the NotebookLM lesson applied to this audience).
- Day-plan generation: 'If you only do one thing for Calculus today' from the student's own notes, offered on the AI landing state.
- A review flow that schedules AI explanations of past wrong answers, turning the dead-end result page into the beginning of the next session.

12. First Thing to Ship

Not the AI. The one-line truth of this app right now is: trust is broken in six places and punishment is everywhere. Fix the broken Send button in group chat, the empty {subject} prompt, the invite-gate review loop, the permanent-delete-with-no-undo gaps, and the missing forward paths - and only then add new features. For a struggling student, a planner that never dead-ends and never punishes IS the feature.

13. Priority Matrix

Order of execution, highest impact first:

- P0 - ship this week:

group-chat Send button; {subject} prompt; skip-link target; course/deadline edit-delete; timed-quiz warning plus result-page forward links; AI Hub and Report in navigation.

- P1 - this month:

remove phantom features or build them; unify terminology; one-primary-action dashboard and Schedule; Settings autosave.

- P2 - next quarter:

grounded Get-Unstuck AI over the student's own notes; 20-minute-day mode; review loop that remembers errors and schedules recovery.